Figures and Summary Table

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Setup

```
# Loading packages
library(haven)
```

```
## Warning: package 'haven' was built under R version 4.5.3
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.3

##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
library(ggplot2)
library(knitr)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.1
## v purrr     1.2.1     v tidyr     1.3.2
## v readr     2.1.6

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter()          masks stats::filter()
## x kableExtra::group_rows() masks dplyr::group_rows()
## x dplyr::lag()             masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
# Imoprting dataset
country.df <- read_csv("Data/Processed/AllCountries_Processed.csv")
```

```
## Rows: 19449 Columns: 25
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (5): Country, HH_ID, Subdivision1, Subdivision2, ElectrificationStatus
## dbl (16): UrbanOrRural, HouseholdSize, DwellingType, DwellingOwnership, Dwel...
## lgl  (4): GridConnectionFeeUSD, InternalWiringFeeUSD, ElectricityBillPriceUS...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Summary table

```r
# Selecting relevant numeric variables
numeric_vars <- c(
  "HouseholdSize",
  "DwellingRoomsNumber",
  "NumYearsGridConnected",
  "GridConnectionFeeUSD",
  "InternalWiringFeeUSD",
  "ElectricityBillPriceUSD",
  "OutageDurationHours",
  "ElectricityConsumptionkWh",
  "EstimateOfElectricitySpendingUSD"
)

# Sorting and calculating descriptive statistics
summary_table <- country.df %>%
  summarise(across(
    all_of(numeric_vars),
    list(
      N = ~sum(!is.na(.)),
      Missing = ~sum(is.na(.)),
      Mean = ~mean(., na.rm=TRUE),
      Median = ~median(., na.rm=TRUE),
      SD = ~sd(., na.rm=TRUE),
      Min = ~min(., na.rm=TRUE),
```

```
      Max = ~max(., na.rm=TRUE)
    )
  )) %>%
  pivot_longer(everything(),
               names_to = c("Variable", ".value"),
               names_pattern = "^(.*)_(N|Missing|Mean|Median|SD|Min|Max)$")
```

```
## Warning: There were 8 warnings in `summarise()`.
## The first warning was:
## i In argument: `across(...)`.
## Caused by warning in `min()`:
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 7 remaining warnings.
```

```
kable(
  summary_table,
  caption = "Descriptive Statistics of Household-Level Variables",
  col.names = c(
    "Variable", "Valid Observations", "Missing Values",
    "Mean", "Median", "SD", "Min", "Max"
  )
)
```

Table 1: Descriptive Statistics of Household-Level Variables

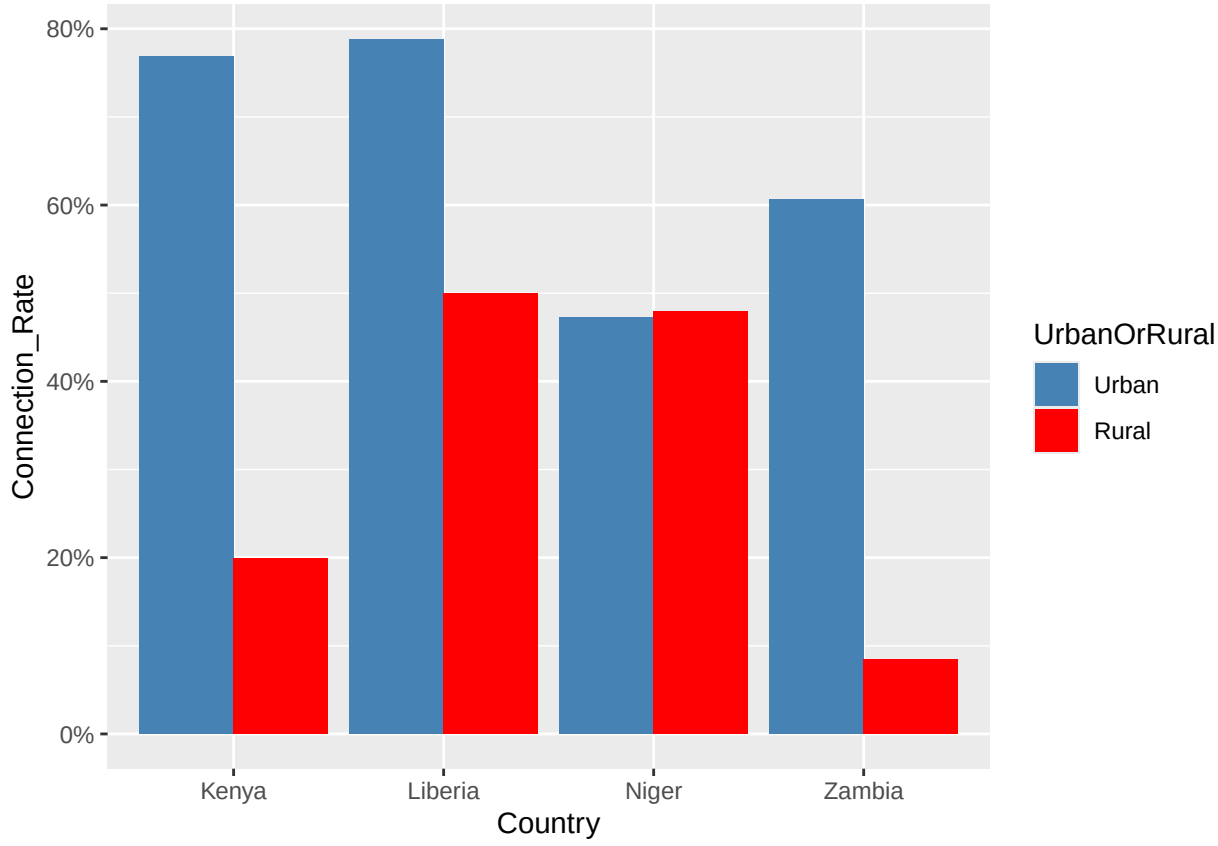
Variable	Valid Observations	Missing Values	Mean	Median	SD	Min	Max
HouseholdSize	12330	7119	4.777372	4.00	2.555887	1	23
DwellingRoomsNumber	15524	3925	3.364275	3.00	23.080112	0	2000
NumYearsGridConnected	7355	12094	353.009517	4.00	2542.548399	0	100000
GridConnectionFeeUSD	0	19449	NaN	NA	NA	Inf	-Inf
InternalWiringFeeUSD	0	19449	NaN	NA	NA	Inf	-Inf
ElectricityBillPriceUSD	0	19449	NaN	NA	NA	Inf	-Inf
OutageDurationHours	6148	13301	235.619226	10.00	375.462276	-8	3300
ElectricityConsumptionkWh	7922	11527	145.693412	4.65	1721.532154	-8	67115
EstimateOfElectricitySpendingUSD	0	19449	NaN	NA	NA	Inf	-Inf

Figure: Urban-rural divide

```
urban_rural <- country.df %>%
  filter(!is.na(UrbanOrRural),
         GridConnected %in% c(1,2),
         UrbanOrRural %in% c(1,2)) %>%
  mutate(
    UrbanOrRural = factor(UrbanOrRural,
                          levels = c(1,2),
                          labels = c("Urban", "Rural"))
  ) %>%
  group_by(Country, UrbanOrRural) %>%
```

```
  summarise(
    Connection_Rate = mean(GridConnected == 1),
    .groups = "drop"
  )

ggplot(urban_rural, aes(x=Country, y=Connection_Rate,
                        fill=UrbanOrRural)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent) + # presenting y axis as %
  scale_fill_manual(
    values = c(
      "Urban" = "steelblue",
      "Rural" = "red"
    )
  )
```




```
  labs(
    title = "Urban vs Rural Comparison of Electricity Access",
    x = "Country",
    y = "Connection Rate (%)",
    fill = "Urban or Rural"
  ) +
    theme_minimal()
```

```
## NULL
```

###Context: This figure compares household electricity connection rates between urban and rural areas across four countries: Kenya, Liberia, Niger, and Zambia. The chart excludes Rwanda because the dataset does not contain any valid urban-rural classifications for its 3,295 household observations, which is a significant limitation.

###Findings: The most notable finding is the substantial urban-rural divide in electricity access in Kenya and Zambia. The difference is approximately 57% in Kenya and 52% in Zambia. Liberia shows a moderate gap of approximately 29%. In contrast, Niger has almost no difference between urban and rural connection rates. Overall, this figure suggests that geographic location is a key factor in determining electricity access.

###Robustness Check:

```
robust_data <- country.df %>%
  filter(
    UrbanOrRural %in% c(1, 2),
    GridConnected %in% c(1, 2)
  ) %>%
  mutate(
    UrbanOrRural = factor(
      UrbanOrRural,
      levels = c(1, 2),
      labels = c("Urban", "Rural")
    ),
    Connected = as.numeric(GridConnected == 1)
  )

# Calculate connection rates for each country
country_rates <- robust_data %>%
  group_by(Country, UrbanOrRural) %>%
  summarise(
    Connection_Rate = mean(Connected) * 100,
    .groups = "drop"
  )

# Calculate equal-weighted average across countries
equal_weighted <- country_rates %>%
  group_by(UrbanOrRural) %>%
  summarise(
    Connection_Rate = mean(Connection_Rate),
    .groups = "drop"
  )

# Calculate pooled household-level rates
pooled_rates <- robust_data %>%
  group_by(UrbanOrRural) %>%
  summarise(
    Connection_Rate = mean(Connected) * 100,
    .groups = "drop"
  )

# Compare the two methods
robustness_table <- bind_rows(
  pooled_rates %>% mutate(Method = "Pooled"),
  equal_weighted %>% mutate(Method = "Equal-weighted")
) %>%
```

```r
  select(Method, UrbanOrRural, Connection_Rate) %>%
  pivot_wider(
    names_from = UrbanOrRural,
    values_from = Connection_Rate
  ) %>%
  mutate(Gap = Urban - Rural)


# Display results
kable(
  robustness_table,
  digits = 1,
  col.names = c(
    "Method",
    "Urban Connection Rate (%)",
    "Rural Connection Rate (%)",
    "Urban-Rural Gap (percentage points)"
  ),
  caption = "Robustness Check: Pooled vs. Equal-Weighted Connection Rates"
)
```

Table 2: Robustness Check: Pooled vs. Equal-Weighted Connection Rates

Method	Urban Connection Rate (%)	Rural Connection Rate (%)	Urban-Rural Gap (percentage points)
Pooled	61.3	24.5	36.8
Equal-weighted	65.9	31.6	34.3

As a robustness check, I compared pooled household-level connection rates with equal-weighted averages of country-specific rates. Under the pooled approach, the urban connection rate was 61.3%, compared with 24.5% for rural households, representing a gap of around 36.8 points. When each country was given equal weight, the average urban electricity connection rates increased to 65.9%, while the rural connection rate increased to 31.6%, narrowing the gap to 34.4 points. Although the overall connection rates differ depending on the method, the disparity remains substantial under both approaches. This suggests that the finding is not driven solely by differences in country sample sizes.